SEMESTER PROJECT: COMPLEX ENGINEERING PROBLEM

COURSE TITLE: HUMAN FACTORS ENGINEERING

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Undertaking

We certify that project work titled ***"Posture Analysis using Different Methods (REBA, RULA & NIOSH)"*** is our own work. No portion of the work presented in this project has been submitted in support of another award or qualification at this either institution or elsewhere. Where material has been used from other sources, it has been properly acknowledged and referred.

Acknowledgment

We have taken efforts in this project. However, it would not have been possible without the kind support and help of our course instructor. I am highly indebted to **Engr. Awais Islam** for their guidance and constant supervision as well as for providing necessary information regarding the project & for their support in completing the project. I would like to express my gratitude towards my parents & member of UET Taxila for their kind co-operation and encouragement, which help me in completion of this project.

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EXECUTIVE SUMMARY

Neck and upper back pains are often attributed to poor posture. For most people, how do we tell if our posture needs attention? **RULA** is a postural targeting method for estimating the risks of work-related upper limb disorders. The results of the test is a Rapid Upper Limb Assessment score. It gives you a fair idea of where you stand with your posture.

REBA provides a quick and easy measure to assess a variety of working postures for risk of WMSDs. It divides the body into sections to be coded independently, according to movement planes and offers a scoring system for muscle activity throughout the entire body, stagnantly, dynamically, fast changing or in an unsteady way and where manual handling may happen which is referred to as a coupling score as it is significant in the loads handling but may not always be using the hands. REBA also gives an action level with a sign of importance and requires minor equipment. It is the best method matched to the needs of occupational safety and health practitioners- and related professions-who have limited time and resources.

NIOSH Lifting Equation, is used for calculating injure-free lifting capabilities for workers who perform two-handed manual lifting tasks. The equation was designed to provide a limit beyond which there would be a need to take ergonomic measures to curtail the risks. The lifting equation is applicable to two-handed lifting tasks. One- and two-handed repetitive lifting tasks form a part of many occupations.

INTRODUCTION:



Ergonomic assessment of Work-Related Musculoskeletal Disorders (WMSDs) involves the evaluation of risk of developing a range of disorders to muscles, nerves and joints, primarily to the upper limb and low back, associated with occupational tasks.

These disorders are caused by different risk factors' interactions resulting from several factors, which can be categorized into individual, psychosocial and physical factors.

Observational and instrument based techniques are proposed in research to provide a quantitative measure for the degree of discomfort and postural strain caused by different body positions.

Observational techniques includes Ovako Working Posture Assessment System (OWAS), Posture, Activity, Tools and Handling (PATH), Quick Exposure Check (QEC), Rapid Upper Limb Assessment (RULA), American Conference of Governmental Industrial Hygienists Threshold Limit Value (ACGHI TLV), Strain Index (SI), Occupational Repetitive Actions (OCRA), NIOSH Lifting Equation, Rapid Entire Body Assessment (REBA).

LITERATURE REVIEW:

Workplaces traditionally have been designed to move products or support machines efficiently. Since people have always seemed so adaptable, how they fit into the workplace has received less attention. The increasing number of injuries caused by repetitive motion, excessive force and awkward postures, ergonomics has become a critical factor in workplace safety. Ergonomics is a combination of the words ergo, a Greek word meaning "work" and nomics, meaning "study" - the study of work. A branch of science that is concerned with the achievement of optimal relationships between workers and their work environment.

In larger scope ergonomics examines human behavioural, psychological, and physiological capabilities and limitations. Professionals in the field of ergonomics normally will design new work environments or modify established work environments based on the studies on the human capabilities and limitation. Ergonomics normally are known to be related to human and their job. MSDs, or musculoskeletal disorders, are injuries and disorders of the soft tissues (muscles, tendons, ligaments, joints, and cartilage) and nervous system. These disorders are not typically the result of a distinctive, singular event, but are more gradual in their development. Thus, MSDs are cumulative-type injuries. It is essential to understand just what a risk factor is, or rather is not. A risk factor itself is not necessarily a causation factor for any particular MSD. They can affect nearly

all tissues, including the nerves and tendon sheaths, and most frequently involve the arms and back.

Three categories of risk factors are identified which are biomechanical exposures, psychosocial stressors and individual risk factors. Biomechanical exposures include factors such as poorly designed workplaces and biomechanical exposures such as repetitive motion, high forces and deviations from neutral body alignments. Psychosocial stressors at work include factors such as high-perceived workplaces stress, low-perceived social support, low perceived job control, and time pressure. Individual factors include gender (female), age, negative stress reactions-especially stomach reactions, and unsatisfactory leisure time or additional domestic workload.

The study will include the ergonomics risk factors in relation of human and their nature of work. Based on the literature, the most significant ergonomics risk factors are awkward posture in handling job task, force and repetition of specific movement including vibration. Other ergonomics risk factor includes uncomfortable static position, contact stress of muscles and tendon and also extreme temperature condition. The Ergonomics Risk Factors (ERF) that are used in this study are listed below:

i) Awkward Posture

ii) Force

iii) Repetition

iv) Vibration

v) Static Loading

PROBLEM STATEMENT

- ***To analyze the given posture using three different methods; REBA, RULA and NIOSH Lifting Equation.***



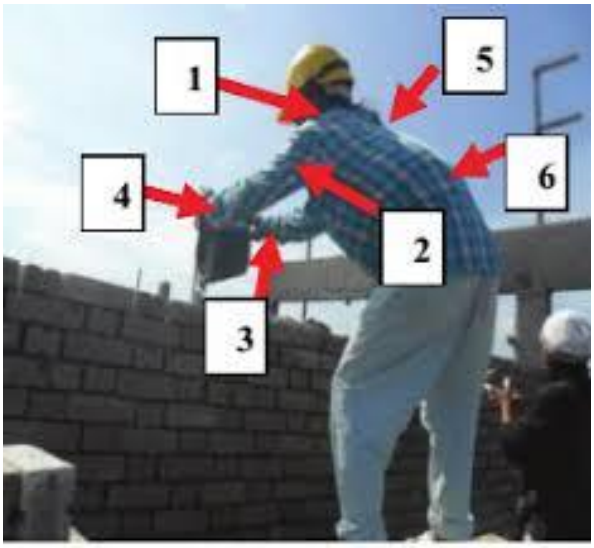


Figure 1: Posture of a worker constructing a wall

OBJECTIVES

To maximize productivity while lowering the risk of Musculoskeletal Disorders (MSDs). MSDs develop as a result of long term exposure to a combination of ergonomic risk factors such as repetition, high forces and awkward postures.

POSTURE ANALYSIS USING RULA METHOD

The **Rapid Upper Limb Assessment (RULA)** is an ergonomics based workplace risk assessment tool that allows you to calculate the risk of musculoskeletal loading within the upper limbs and neck. RULA is easy and quick to use and does require expensive equipment to complete.

- RULA is designed to be a rapid tool.
- RULA is intended to be part of a broader ergonomics survey.
- RULA is a screening tool that assesses biomechanical and postural loading on the whole body with particular attention to the neck, trunk and upper limbs.
- A RULA assessment requires little time to complete and the scores generated fit into an action list which indicates the level of intervention required.

RULA Advantages:

- Good for upper limb tasks.
- Measuring musculoskeletal risks.
- Evaluating outcomes such as suitability of equipment and productivity.
- Great for educating workers around high risk postures.
- Good for tasks that are mostly static.
- Showing an objective number before and after interventions (useful for making the business case when presenting to colleagues or management).
- Reasonable accurate and reliable.
- Quick and easy to use.
- Has shown to improve productivity.
- Useful for presenting info to management (gives a number; is 'objective')
- Can compare pre- post- intervention
- Reasonably accurate and reliable
- Can demonstrate relationship between score and regional pain.

RULA Disadvantages:

- Tasks that involve the whole body.
- Wrist posture does not take into account position of function which has some wrist extension.
- Tasks that are varied.
- Does not take into account organizational and psychological factors.
- Occasional discrepancy of scores when limbs are at borders of two ranges.
- Difficult to catch with photograph. (Video is ideal)
- Limited to accessing one moment in time. Have to choose most extreme posture.
- May require several RULA assessments for one task.
- Not so good for tasks that involve the whole body and/or are varied.

The RULA worksheet is divided into two body segment sections on the labeled A and B. Section A (left side) covers the arm and wrist. Section B (right side) covers the neck, trunk and legs. This segmenting of the worksheet ensures that any awkward or constrained postures

of the neck, trunk or legs which might influence the postures of the arms and wrist are included in the assessment.

Methods selection for this posture (CEP attribute-5)

RULA METHOD

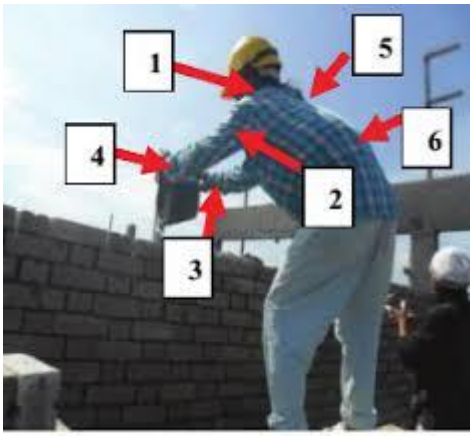


Figure 3:



Figure 2

A: Right arm and wrist analysis

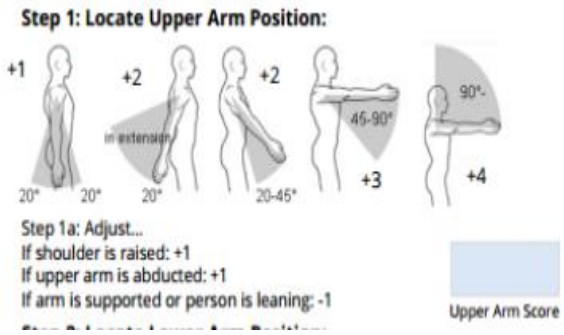
- **Step 1: Locate Upper arm position**

Adjust = +3

Shoulder is raised = +1

Person is leaning= -1

Score = +4

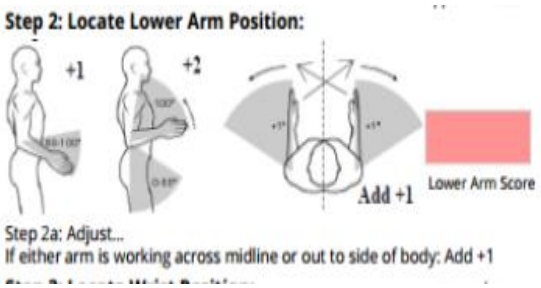


- **Step 2: Locate lower arm position**

Adjust = +2

Arm is working out to side of the body = +1

Score = +3



- **Step 3: Locate Wrist Position**

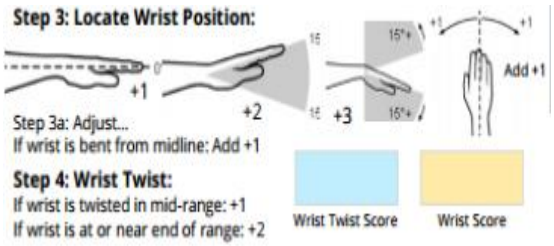
Adjust = +3

Across movement of hands = +1

Score = +4

- **Step 4: Wrist Twist**

If wrist is at or near end of range = +2



- **Step 5: Look up Posture score in Table A**

Upper arm score = 4

Wrist Score = 4

Lower arm score = 3

Wrist twist = 2

Posture Score A = 6

Scores

Table A		Wrist Score							
		1		2		3		4	
Upper Arm	Lower Arm	Wrist Twist		Wrist Twist		Wrist Twist		Wrist Twist	
		1	2	1	2	1	2	1	2
1	1	1	2	2	2	2	3	3	3
	2	2	2	2	2	3	3	3	3
	3	2	3	3	3	3	3	4	4
2	1	2	3	3	3	3	4	4	4
	2	3	3	3	3	3	4	4	4
	3	3	4	4	4	4	4	5	5
3	1	3	3	4	4	4	4	5	5
	2	3	4	4	4	4	4	5	5
	3	4	4	4	4	4	5	5	5
4	1	4	4	4	4	4	5	5	5
	2	4	4	4	4	4	5	5	5
	3	4	4	4	5	5	5	6	6
5	1	5	5	5	5	5	6	6	7
	2	5	6	6	6	6	7	7	7
	3	6	6	6	7	7	7	7	8
6	1	7	7	7	7	7	8	8	9
	2	8	8	8	8	8	9	9	9
	3	9	9	9	9	9	9	9	9

- **Step 6: Add muscle use score**

As this action is repeated 4X per minute so Adding ***+1***

- **Step 7: Add force/load score**

Load = 4.5 lbs and action is repeated so score = ***+2***

- **Step 8: Wrist and Arm score**

Posture score A + Muscle Score + Force/load Score = Wrist and Arm Score

6+1+2= 9

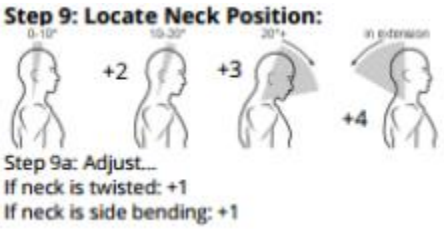
B. Neck, trunk and Leg Analysis

- **Step 9: Locate neck position**

Adjust = +3

Neck is twisted +1

Score = 4

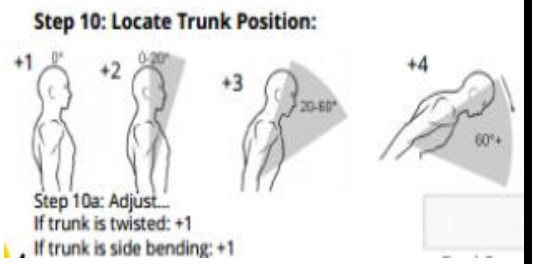


- **Step 10: Locate trunk Position**

Adjust = +3

Twisted Trunk = +1

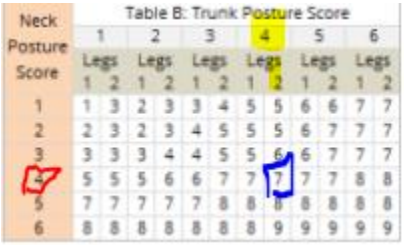
Score = 4



- **Step 11: Legs**

Legs and feet are not supported so score = +2

- **Step 12: Look up posture score in Table B.**



According to table:

Score = 7

- **Step 13: Add Muscle use score.**

Action repeated for more than 4x, so score = +1

- **Step 14: Add force/load score.**

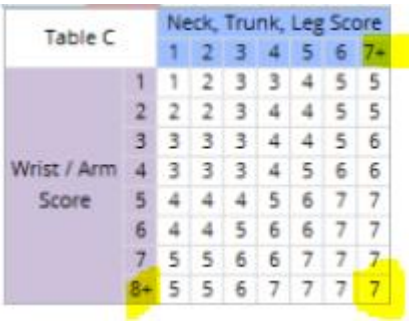
Load = 4.5 lbs and action is repeated so score = +2

- **Step 15: Add all scores.**

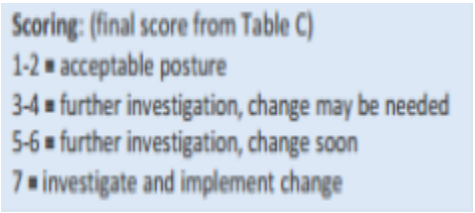
Posture score B + Muscle Score + Force/load Score = Neck, trunk and leg score

7+1+2=10

- **Step 16: Find value in Table C.**



The value from the table C =7



RULA score =7

According to this, 7 is high risk factor which needs investigation and further changes.

Rapid Upper Limb Assessment Levels:

It works as a function of the degree of injury risk.

- **Action Level 1** – RULA score **1-2** means that the person is working in the best posture.
- **Action Level 2** – RULA score **3-4** means that the person is working in a posture that could present some risk of injury from their work posture. This score most likely is the result of one part of the body being in a deviated and awkward position. Investigate the reasons and correct.

- **Action Level 3** – RULA score **5-6** means that the person is working in a poor posture with a risk of injury from their work posture.
- **Action Level 4** – RULA score **7-8** means that the person is working in the worst posture with an immediate risk of injury from their work posture. Investigate the reasons and change them in near future to prevent an injury.

POSTURE ANALYSIS USING REBA METHOD

REBA METHOD:

This ergonomic assessment tool uses a systematic process to evaluate whole body postural MSD and risks associated with job tasks. A single page worksheet is used to evaluate required or selected body posture, forceful exertions, type of movement or action, repetition, and coupling.

BENEFITS OF REBA:

- One of the benefits of REBA is that it evaluates different body parts: upper Limbs (arm, forearm and wrist), lower extremities, trunk and neck.
- It is a useful method to identify the forced postures adopted by workers to thus develop improvement measures if necessary.

ADVANTAGES OF REBA:

- The main advantages of the REBA method are:
- The cost-effectiveness ratio is good.
- It is easy to apply. Pen and paper are enough for data collection; however, there are computer applications that speed up/facilitate its use.
- The most conflictive ergonomic aspects are identified from the individual score obtained after assessing each part of the body.

LIMITATIONS OF REBA:

The main limitations are:

- It only allows the analysis of individual postures. It is not possible to analyse a set or sequence of postures.
- Task evaluations will depend on the evaluator. Some of the positions adopted may or may not be examined.
- It only measures the effort intensity. The duration of exposure and the frequency of postures throughout the working day are not considered.

The REBA worksheet is divided into two body segment sections on the labeled A and B. Section A (left side) covers the neck, trunk, and leg. Section B (right side) covers the arm and wrist. This

segmenting of the worksheet ensures that any awkward or constrained postures of the neck, trunk or legs which might influence the postures of the arms and wrist are included in the assessment.

REBA METHOD

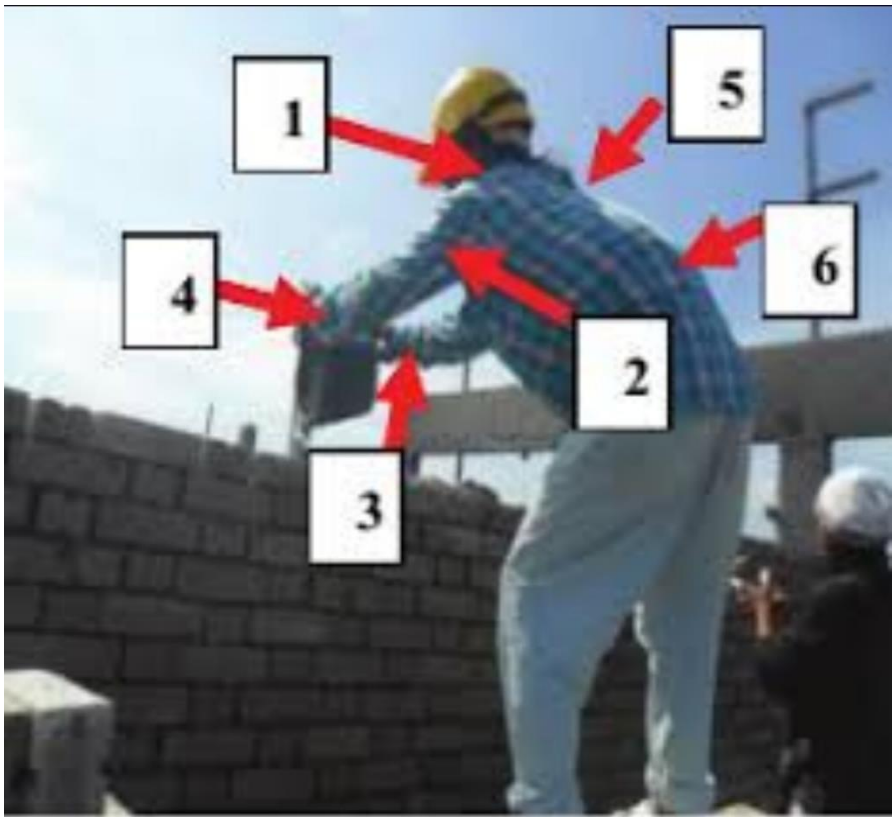


Figure 4

First, we calculate **"Neck, Trunk and Legs"** posture for Group A.

Scores

Table A		Neck											
		1				2				3			
	Legs	1	2	3	4	1	2	3	4	1	2	3	4
Trunk Posture Score	1	1	2	3	4	1	2	3	4	3	3	5	6
	2	2	3	4	5	3	4	5	6	4	5	6	7
	3	2	4	5	6	4	5	6	7	5	6	7	8
	4	3	5	6	7	5	6	7	8	6	7	8	9
	5	4	6	7	8	6	7	8	9	7	8	9	9

STEP#1: LOCATE NECK POSITION

Score is +1

STEP#2: LOCATE TRUNK POSITION

Score is +3

STEP#3: LOCATE LEGS POSITION

Score is +3. (+2 and +1 Angle 30-60 degree)

STEP#4: LOOK UP POSTURE SCORE IN TABLE A

Score found in table A is 5.

STEP#5: ADD FORCE/LOAD SCORE

Load is 4.5 lbs < 11 lbs. Score is +0

STEP#6: SCORE A, FIND ROW IN TABLE C

Add values from step#4 to 5. Score is 5+0 = 5

Now, we calculate **"ARM AND WRIST ANALYSIS"** posture for Group B.

Table B		Lower Arm					
		1			2		
	Wrist	1	2	3	1	2	3
Upper Arm Score	1	1	2	2	1	2	3
	2	1	2	3	2	3	4
	3	3	4	5	4	5	5
	4	4	5	5	5	6	7
	5	6	7	8	7	8	8
	6	7	8	8	8	9	9

STEP#7: LOCATE UPPER ARM POSITION

Score is +4 (+3 as angle is 45-90 degrees and +1 as upper arm is abducted)

STEP#8: LOCATE LOWER ARM POSITION

Score is +2

STEP#9: LOCATE WRIST POSITION

Score is +2 (wrist is twisted)

STEP#10: LOOK UP POSTURE SCORE IN TABLE B

Score found in table B is 6.

STEP#11: ADD COUPLING SCORE

Score is +3 (No handles, awkward, unsafe with any body part)

STEP#12: SCORE B, FIND COLUMN IN TABLE C

Add all the values from step#10 to 11. Score is 6+3 = 9

Now, we calculate **"SCORE"** in Table C

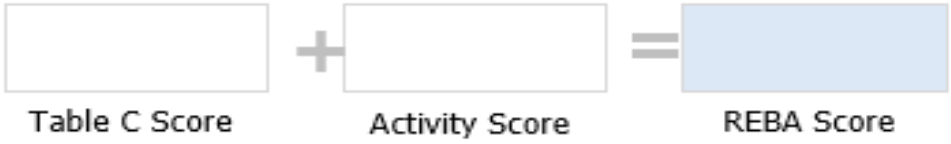
Score A	Table C											
	Score B											
	1	2	3	4	5	6	7	8	9	10	11	12
1	1	1	1	2	3	3	4	5	6	7	7	7
2	1	2	2	3	4	4	5	6	6	7	7	8
3	2	3	3	3	4	5	6	7	7	8	8	8
4	3	4	4	4	5	6	7	8	8	9	9	9
5	4	4	4	5	6	7	8	8	9	9	9	9
6	6	6	6	7	8	8	9	9	10	10	10	10
7	7	7	7	8	9	9	9	10	10	11	11	11
8	8	8	8	9	10	10	10	10	10	11	11	11
9	9	9	9	10	10	10	11	11	11	12	12	12
10	10	10	10	11	11	11	11	12	12	12	12	12
11	11	11	11	11	12	12	12	12	12	12	12	12

Score is 9.

STEP#13: ACTIVITY SCORE

Activity score from Table A and B, gives 9 on Table C and+1 is added for large action changes in posture. So, final score is 10.

FINAL REBA SCORE



Final score is 10.

Score	Level of MSD Risk
1	negligible risk, no action required
2-3	low risk, change may be needed
4-7	medium risk, further investigation, change soon
8-10	high risk, investigate and implement change
11+	very high risk, implement change

It shows high risk and need changes to implement for reduction of MSD risk.

DIFFERENCE BETWEEN RULA AND REBA:

Familiarities of Issues and difference *(CEP attribute-6)*

- REBA assessment is suitable for whole body evaluation and best for both static and dynamic works. There are five levels of actions to indicate the obtained scores. RULA assessment is more towards upper side of the body. It is best for sedentary and seated works and there are four levels of actions to indicate the obtained scores.
- REBA was improved in the definition of the neutral postures and leg postures from those of RULA. While RULA defined the neutral position of the wrist, neck and trunk with 0° of corresponding joint motion angle.

- REBA adapts better than RULA to highly varied workstations. This can be attributed to the development of RULA within a specific research context that makes it unreliable when applied in a different context.

NIOSH LIFTING EQUATION

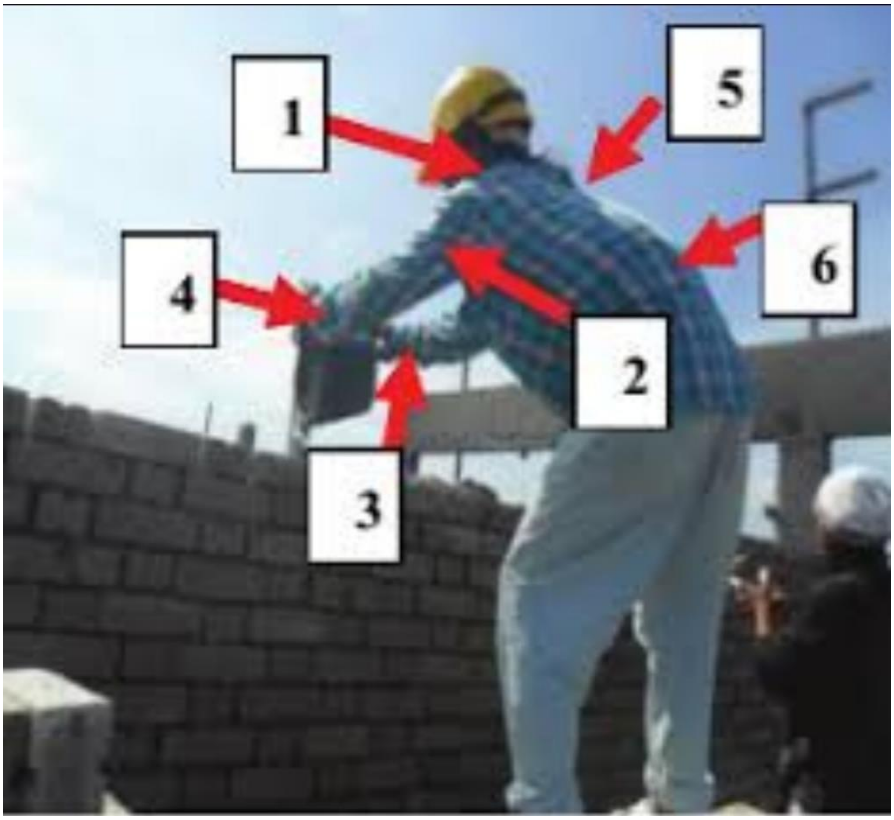


Figure 5

NIOSH LIFTING EQUATION:

The Revised NIOSH Lifting Equation is a tool used by occupational health and safety professionals to assess the manual material handling risks associated with lifting and lowering tasks in the workplace.

ADVANTAGE OF NIOSH:

NIOSH has developed a mathematical model that helps predict the risk of lifting-related injuries. The lifting equation defines a recommended weight limit (RWL) for specific lifting tasks that most workers could perform over an eight-hour day without increasing the risk of developing lower back pain.

Limitations of the NIOSH lifting equation:

- It only applies to two-handed manual lifting tasks, from a standing position.
- The calculations are based on the assumption that other manual material handing activities are minimal (less than about 10% of worker activity)

NIOSH Lifting Equation:

The Equation:

$$RWL = 51 \left[\frac{10}{H}\right] \left[1 - (.0075 \times |V-30|)\right] \left[.82 + \frac{1.8}{D}\right] \left[F\right] \left[1 - .0032A\right] \left[C\right]$$

Task variables needed to calculate the RWL:

- H = Horizontal location of the object relative to the body
- V = Vertical location of the object relative to the floor
- D = Distance the object is moved vertically
- A = Asymmetry angle or twisting requirement
- F = Frequency and duration of lifting activity
- C = Coupling or quality of the workers grip on the object

RWL:

The Recommended Weight Limit (RWL) of the NIOSH lifting equation is the product of the load constant and six multipliers, which can be expressed as follows:

RWL = LC • HM • VM • DM • AM • FM • CM

Where, LC—load constant HM—horizontal multiplier, VM—vertical multiplier, DM—distance multiplier, AM—asymmetry multiplier, FM—frequency multiplier, CM—coupling multiplier.

Additional task variables needed to calculate LI:

- Average weight of the objects lifted
- Maximum weight of the objects lifted

For LI:

$$\text{Lifting Index (LI)} = \frac{\text{actual load}}{\text{RWL}}$$

Frequency Multiplier Table (FM)

Frequency	Work Duration (Continuous)					
	<8 HRS		<2 HRS		<1 HOUR	
Lifts/min	V<30	V>30	V<30	V>30	V<30	V>30
0.2	.85	.85	.95	.95	1.00	1.00
0.5	.81	.81	.92	.92	.97	.97
1	.75	.75	.88	.88	.94	.94
2	.65	.65	.84	.84	.91	.91
3	.55	.55	.79	.79	.88	.88
4	.45	.45	.72	.72	.84	.84
5	.35	.35	.60	.60	.80	.80
6	.27	.27	.50	.50	.75	.75
7	.22	.22	.42	.42	.70	.70
8	.18	.18	.35	.35	.60	.60
9	.00	.15	.30	.30	.52	.52
10	.00	.13	.26	.26	.45	.45
11	.00	.00	.00	.23	.41	.41
12	.00	.00	.00	.21	.37	.37
13	.00	.00	.00	.00	.00	.34
14	.00	.00	.00	.00	.00	.31
15	.00	.00	.00	.00	.00	.28
>15	.00	.00	.00	.00	.00	.00

Hand—to—Container Coupling Classification

Couplings	V<30 inch	V>30 inch
	Coupling Modifiers	
Good	1.00	1.00
Fair	0.95	1.00
Poor	0.90	0.90

CALCULATION:

H = 1.5ft = 18inches

V(origin) = 4ft = 48inches

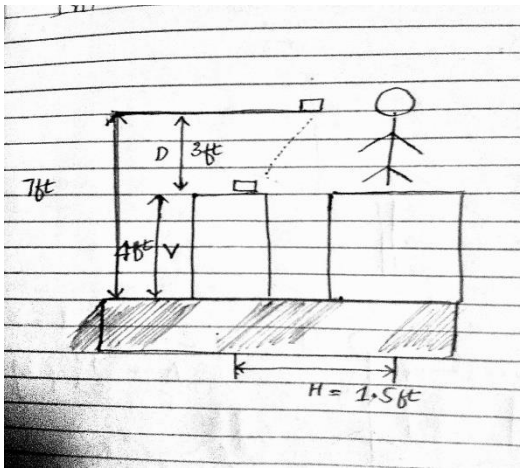
D = 3ft = 36inches

A = 90 degrees

F = 5lift/min

C = 1 (good)

Actual Load = (2kg)



Put the values in the equation:

RWL = 51[10/18] [1- (.0075*|48-30|] [.82+1.8/36] [5] [1-(.0032*90)] [3]

= (28.33) (0.865) (0.87) (0.712) (0.35) (1)

= 5.3218

LI = Actual load/RWL = 2/5.3218 =0.37644

V from Destination

H = 1.5ft = 18inches

V = 7ft = 84inches

D = 3ft = 36inches

A = 90 degrees

F = 5lift/min= 0.35

C = 1 (good)

Actual Load = 2kg

Put the values in the equation:

RWL = 51[10/18] [1- (.0075*|84-30|] [.82+1.8/36] [5] [1-(.0032*90)] [1]

= (28.33) (0.595) (0.87) (0.712) (0.35) (1)

= 3.656

LI = Actual load/RWL = 2/3.656 = 0.0547

RECOMMENDATIONS

Analysis Required *(Attribute of CEP -2)*

- **RULA** result recommended changing from bending work posture using the old tool into standing work posture. The old work posture (P0) showed RULA grand score of 7 with high risk level, while the new work posture (P1) showed RULA grand score of 3 with low risk level.
- The **REBA** method is mainly used for the analysis of forced postures. It is not useful for the evaluation of repetitive movements. When using REBA, only the right or left side is assessed at a time. After interviewing and observing the worker the evaluator can determine if only one arm should be evaluated, or if an assessment is needed for both sides. The REBA method is not recommended for evaluating manual handling tasks.
- **Recommended weight limit according to NIOSH lifting equation:** The maximum weight to be lifted with two hands, under ideal conditions, is 51 pounds. The RWL is based on six variables that reduce the maximum weight to be lifted to less than 51 pounds.



Figure 6: Support across arms helps them

- As in figure, it is shown that the worker's back is bending and this posture creates a difficulty in order to work. So, it is recommended that he should work according to the proper height of wall as shown in the Figure 7 (a).
- Second, worker bears a lot of load of bricks in order to make a wall. So, there must be a place on which all the bricks are easily handled by the worker. It also reduces the time of work.
- Third is that the worker is not working in safe condition. His legs are not supported to each other. There is a risk of falling down or feet injury. So, he should stand according to appropriate height and work on safe place as shown in Figure 6.



Figure 7 (a): Recommended Safety and suggestions



Figure 7 (b)

CONCLUSION:

Conclusion (Attribute of CEP-7)

RULA is a popular ergonomic assessment method to identify risk factors of the posture, force and muscle risk factors for work related musculoskeletal disorder. RULA considers biomechanical and postural load requirements of job tasks/demands on the neck, trunk and upper extremities.

REBA is one of the most popular and widely used observational ergonomic assessment tools in various industries and services. Being an observational assessment tool, REBA is a subjective method; it lacks detail and precision and covers three important risk factors: Force, repetition and posture. Future work is needed to support the predictive and concurrent validity and reliability of the method.

The NIOSH lifting Equation has its application in a variety of fields to reduce the Low Back Pain caused by manual lifting tasks at workplaces. The study based on these applications is being used to know effectiveness of lifting tasks in order to ensure safe working practices of workers.

REFERENCES:

1. Human Factors in Engineering and Design" by E J McCormick.
2. Textbook of Work Physiology" by O P Astrand and R Kaare.
3. The Body at Work: Biological Ergonomics" by W T Singleton.

Sr. No	**Attribute**	**Description**
1.	Range of conflicting requirements	Design should satisfy the both 5th to 95th percentile of students according to their anthropometry.
2.	Depth of analysis required	Data distribution, risk assessment (RULA method, RIBA assignment) and questioners.
3.	Depth Of Knowledge	Includes the knowledge of human ergonomics and Anthropometry.
4.	Extent of stakeholder involvement and level of Conflicting Requirements	Support of departments and students during data collection. Equipment provided by the department for the measurement.
5.	Extent of applicable codes	Organization should provide the ergonomically suitable designed chair for students during lectures.
6.	Consequences	Students will get benefit from the new improved design in terms of comfort, concentrations and in the prevention of musculoskeletal problems. Organizations will use our design and suggestions while installing new chairs.